

IEOR 4404: Simulation

Lecture 8: Large-Sample Properties

Large-Sample Behavior

- We want to investigate the behavior of “averages” over independent and identically distribution (i.i.d.) random variables
- Given i.i.d. r.v.'s $X_1, \dots, X_n$, consider the *sample mean*

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

$\underbrace{E[X]}$

- Question: when n is large, sample mean $\approx$ true mean?
- More questions:
 - What does it mean to be a good approximation?
 - How “large” is large?

Large-Sample Behavior

In Monte Carlo simulation, this is motivated from output analysis, i.e., statistical inference on simulation outputs

The theorems we discuss form the foundation of simulation output analysis

Sample Mean

- Given n i.i.d. data $X_1, \dots, X_n$, we like to estimate $\mu = E[X]$

Sample Mean

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- Mean of $\bar{X}$ is μ : Note: $\bar{X}$ is a random variable

$$E[\bar{X}] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = \mu$$

Sample Mean

- Given n i.i.d. data $X_1, \dots, X_n$, we like to estimate $\mu = E[X]$
- Mean of $\bar{X}$ is μ :

$$E[\bar{X}] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = \mu$$

- Suppose $Var(X) = \sigma^2$. Variance of $\bar{X}$ is σ^2/n :

$$\begin{aligned} Var(\bar{X}) &= Var\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{Var(X_1 + \dots + X_n)}{n^2} = \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

Is it true that $\bar{X} \approx \mu = E(X)$?



• $E[\bar{X}] = \mu$

But, it doesn't imply $\bar{X} \approx \mu$.

• $\text{var}(\bar{X}) = \frac{\sigma^2}{n}$

This implies that, for large n , $\text{var}(\bar{X}) \approx 0$
 $\Rightarrow \bar{X}$ has a very small variability.

Law of Large Numbers

When n is large, the sample mean

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

is close to the true mean μ

Theorem (Law of Large Numbers). Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, as $n \rightarrow \infty$, $P(|\bar{X} - \mu| \text{ is small}) \rightarrow 1$.

Markov's Inequality

Theorem (Markov's Inequality). Let X be a nonnegative random variable with finite mean $E[X]$. Then, for any $a > 0$,

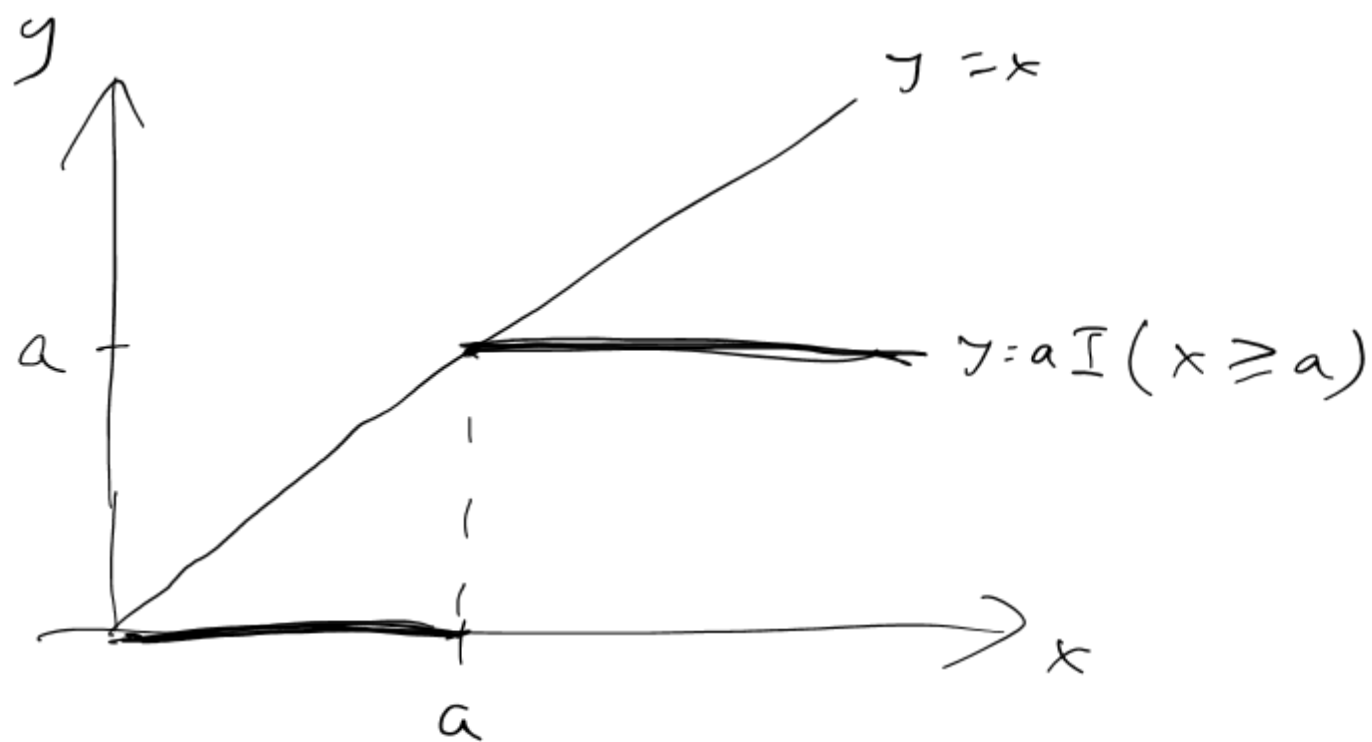
$$P(X \geq a) \leq E[X]/a$$

Proof.

For any $a > 0$, $\mathbf{1}\{X \geq a\} \leq X/a$, where $\mathbf{1}\{X \geq a\} = 1$ if $X \geq a$ and 0 otherwise

So $E[\mathbf{1}\{X \geq a\}] \leq E[X/a]$

or equivalently $P(X \geq a) \leq E[X]/a$



$$a I(x \geq a) \leq x$$

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View x as a random variable,
and take expectation on both sides.

$$E[a I(x \geq a)] \leq E[x]$$

$$\Rightarrow a E[I(x \geq a)] \leq E[x]$$

$$\Rightarrow a p(x \geq a) \leq E[x]$$

Chebyshev's Inequality

Theorem (Chebyshev's Inequality). Let X be a random variable with finite mean μ and finite variance σ^2 . Then for any $k > 0$,

$$P(|X - \mu| \geq k) \leq \sigma^2/k^2$$

Proof. Apply Markov's inequality to the random variable $Y^2 = |X - \mu|^2$

Then $P(|X - \mu| \geq k) = P(Y \geq k) = P(Y^2 \geq k^2) \leq E[Y^2]/k^2 = \sigma^2/k^2$

$$P(\underbrace{|x - \mu|^2}_{\text{non-negative r.v.}} \geq k^2) \leq \frac{E|x - \mu|^2}{k^2} = \frac{\sigma^2}{k^2}$$

view this as

a non-negative r.v.

by Markov inequality.

where $\mu = E[x]$, $\sigma^2 = \text{var}(x)$

Law of Large Numbers

Theorem (Law of Large Numbers) [more precise version]. Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, as $n \rightarrow \infty$,

$$P(|\bar{X} - \mu| \leq \epsilon) \rightarrow 1$$

for any $\epsilon > 0$.

Proof.

Apply Chebyshev's inequality to $\bar{X}$:

Mean of $\bar{X}$ is μ

Variance of $\bar{X}$ is σ^2/n

So, for any $\epsilon > 0$,

$$P(|\bar{X} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Equivalently,

$$P(|\bar{X} - \mu| \leq \epsilon) \rightarrow 1$$

In the Chebyshev inequality,

take $\bar{X}$ as the r.v.

$$P(|\bar{X} - \mu| \geq k) \leq \frac{\text{var}(\bar{X})}{k^2} = \frac{\sigma^2}{k^2 n}$$

$\rightarrow 0$ as $n \rightarrow \infty$

So, when n is big,

$$P(|\bar{X} - \mu| \geq k) \approx 0 \text{ or } P(|\bar{X} - \mu| < k) \approx 1.$$

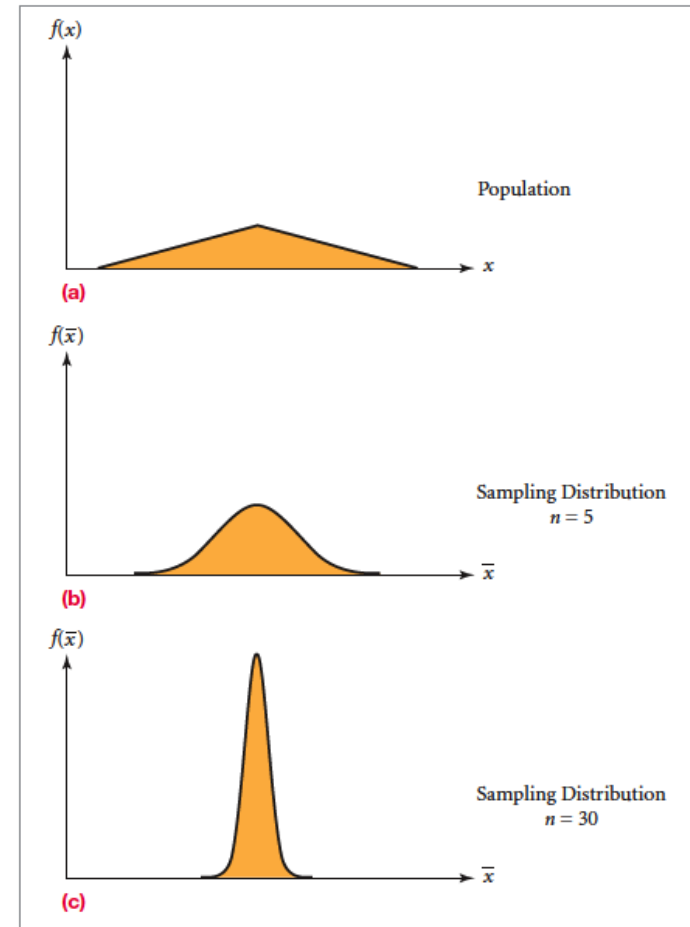
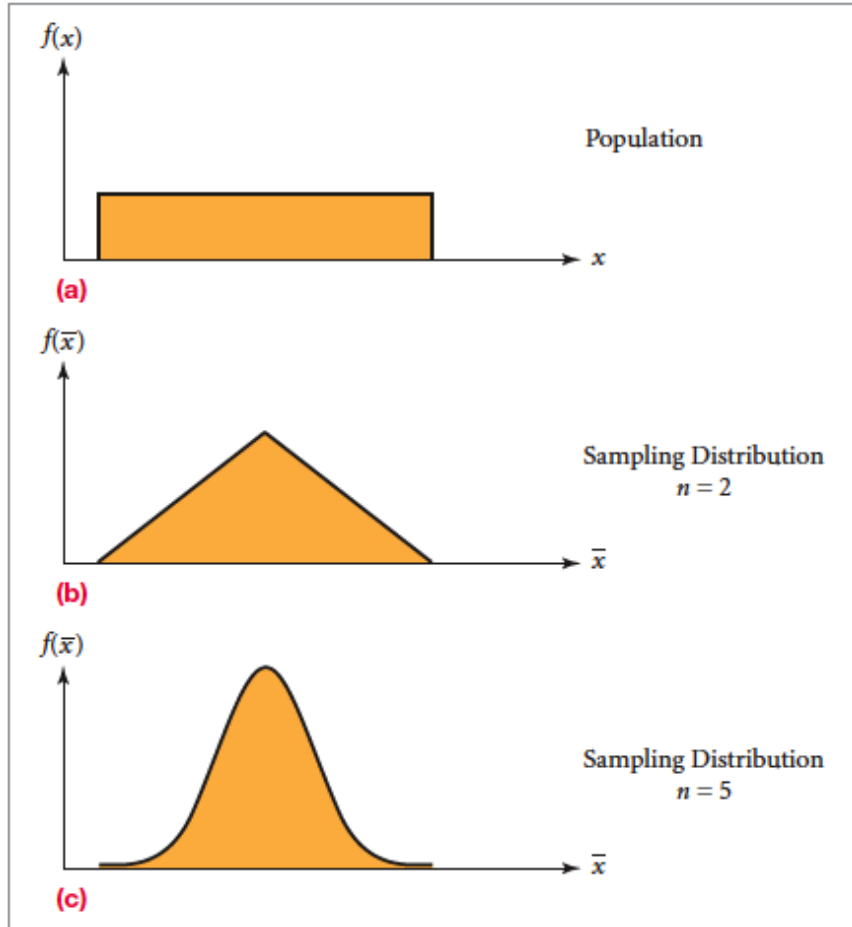
Central Limit Theorem

- From the law of large numbers, we know that the sample mean is a good estimate of the true expectation, *provided* that the sample size is large
- How large is large enough?
- **Theorem (Central Limit Theorem).** Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, for large n , the distribution of $\bar{X}$ is approximately normal with mean μ and variance σ^2/n .

$$\bar{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\text{where } \mu = E[X], \quad \sigma^2 = \text{var}(X)$$

Central Limit Theorem



Central Limit Theorem

- Implications of the central limit theorem:

$$P(\bar{X} \leq x) \approx P(Y \leq x)$$

where $Y \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

- The approximation can also be written as

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq x\right) \approx P(Z \leq x)$$

where $Z \sim N(0,1)$

- A rule of thumb is that the above approximation holds when n is at least 30

Sample vs Sampling Distributions

- The distribution of $\bar{X}$ is called the *sampling distribution*
- The object $\bar{X}$ (often called a statistic) and its distribution depends on n i.i.d. data $X_1, \dots, X_n$
- This distribution relates to, but not the same as, the distribution of any sample point X_i , e.g.,
 - mean of X_i = mean of $\bar{X}$ = μ
 - var of X_i = σ^2 , but var of $\bar{X}$ = σ^2/n
- By the law of large numbers, we expect the sampling distribution of $\bar{X}$ to concentrate around μ
- By the central limit theorem, we know the sampling distribution of $\bar{X}$ is approximately a normal distribution with mean μ and variance σ^2/n